AI-Based Smart Retail Recommendation System for Apparel

Enhancing Customer Experience, Inventory Optimization, and Marketing Intelligence using AI



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Proposal Feedback & Improvements

Student ID	Panel Comment	How we Address the Issue
IT22243812 L.A.D.A.S. Perera	Detecting emotions from CCTV camera is challenging and doing it REALTIME is another challenge	Removed CCTV-based emotion detection. Introduced **GPS-based behaviour tracking** to analyze movement, zone changes, and time spent, enabling real-time staff assistance in a simpler and practical way
IT22284952 Silva T. A	Virtual clothing requires advanced VR/AR expertise	Replaced complex VR/AR with a **2D virtual try-on approach using APIs**, reducing complexity while maintaining core functionality
IT22244598 Methsara W.M	Inventory systems already exist; problem justification is weak	Refocused problem on **limitations of existing systems**, emphasizing AI-based forecasting, stock optimization, and real-time decision support instead of basic inventory tracking
IT22146588 | WIJETUNGA J A R K	Marketing dashboard link to promotions and research contribution is unclear	Refined system to be **fully promotion-focused**: predicts best promotions (Random Forest), analyzes performance, and generates posters (Stable Diffusion). Integrated analytics + AI + content generation into one pipeline

Problem & Impact

The Problem

- Traditional retail systems focus only on **sales and inventory**
- Retailers lack insights into:
  - customer behavior
  - product preferences
  - promotion effectiveness
- Large amounts of retail data remain **unused**
- **Absence on integrated systems**
- **Absence of smart systems for stakeholder decisions and use of manual traditional methods**

Impact

- Poor customer experience
- Ineffective marketing strategies
- Stock shortages and overstock issues
- Reduced sales and business performance
- Making poor decisions based on manual analysis
- No clear view of customer behavior

Users & Stakeholders



Retail Customers

Seek discovery, relevance and a frictionless in-store journey.



Store Staffs & Managers

Need actionable insights to optimise optimise layout, staff and stock in real time.



Marketing Teams

Require behaviour-driven signals to tailor promotions and measure campaign impact.



Business & Supply Chain

Care about inventory flow, cost cost control and reliable demand forecasts.



Design Excellence of the Solution

- Integrates **multiple AI-driven components in one platform**
- Uses:
  - Machine Learning
  - Data Analytics
  - Generative AI
- Provides:
  - Real-time insights
  - Predictive analytics
  - Decision support

Key Strength:

Transforms raw retail data into intelligent business decisions



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Our Proposed System

GPS-Based Customer Behaviour Tracking & Staff Assistance

Tracks customer movement patterns and provides real-time assistance alerts to store staff.

AI-Based Virtual Try-On & Size Recommendation

Allows customers to virtually try products and receive accurate size recommendations using AI.

Smart Inventory & Stock Flow Optimization

Predicts product demand and optimizes inventory to prevent stock shortages and overstock.

Personalized Marketing Intelligence Dashboard

Analyzes customer data to generate targeted promotions and measure campaign performance.

Problem Definition



Made with GAMMA

IT22243812 | L.A.D.A.S. Perera

Specialization: Information Technology



GPS-Based Customer Behaviour Tracking & Real-Time Staff Assistance

Problem Definition & Research Gap

Research Gap

Existing retail tracking systems mainly focus on customer monitoring using CCTV or sensors, with limited real-time staff assistance and privacy-friendly solutions.

Novelty

This research introduces a GPS-based zone tracking system that combines customer behaviour analysis, difficulty detection, real-time staff alerts, and store analytics in one solution.

Solution Overview

Problem

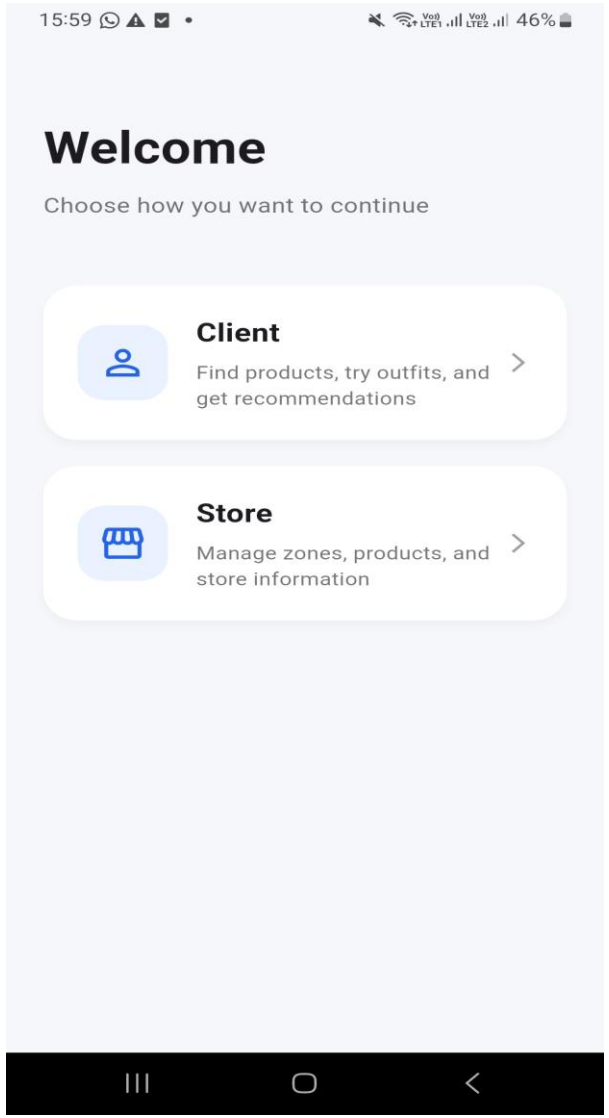
- Difficult to identify customers needing assistance
- Manual monitoring is inefficient
- Real-time observation is challenging

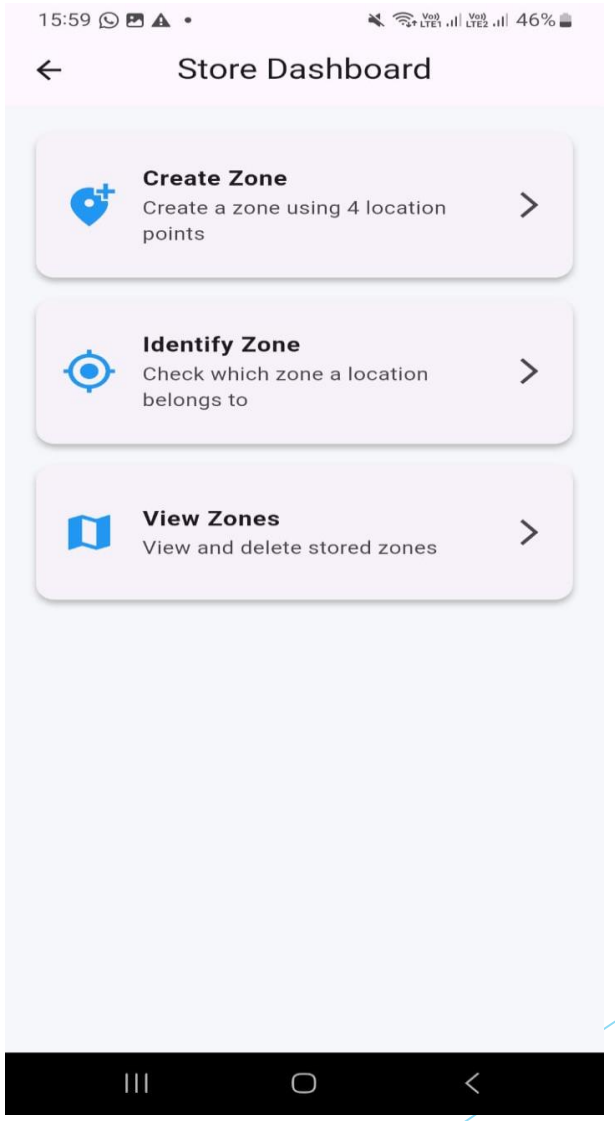
Solution

- GPS-based indoor customer tracking
- Detects confusion, hesitation, and product interest
- Helps staff provide timely assistance

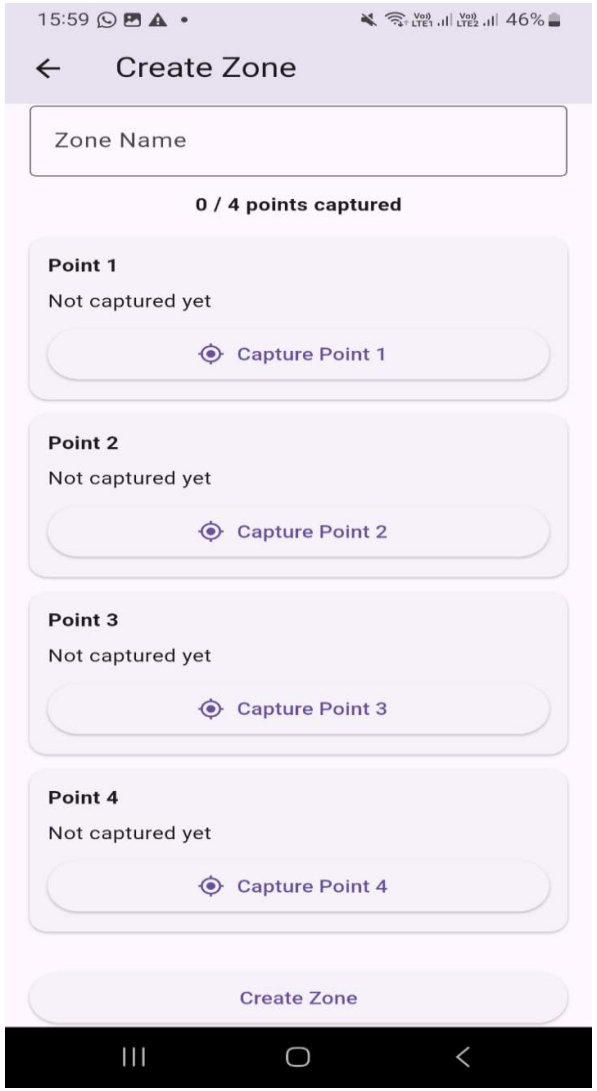
Workflow & Prototype

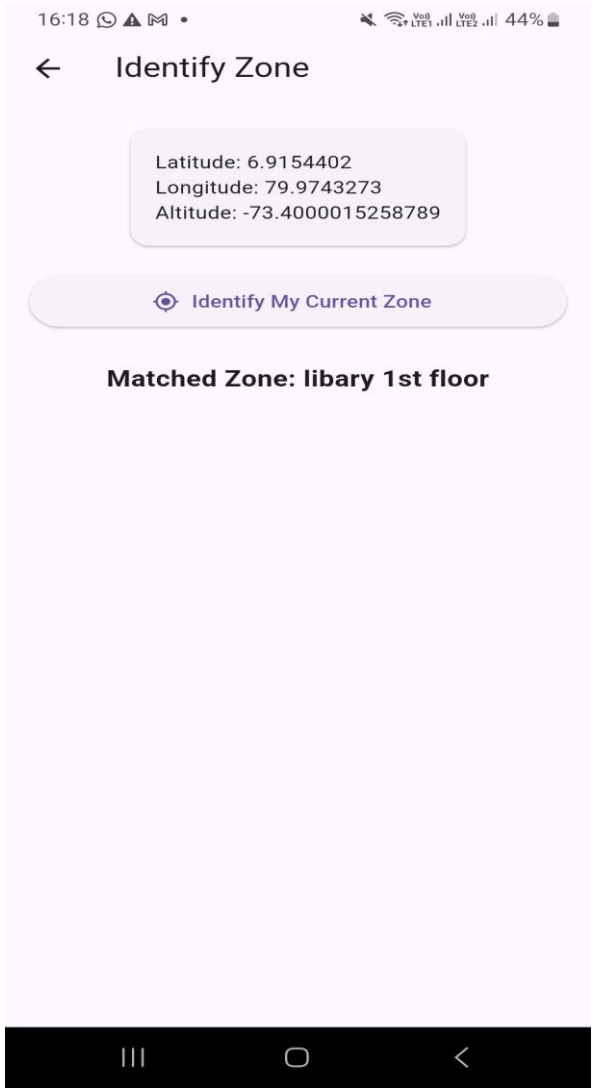
- Divide the store into zones.
- Detect the customer’s current zone using GPS.
- Track customer movements inside the store.
- Analyze customer behaviour patterns.
- Identify possible customer difficulties.
- Send instant alerts to staff.
- Show store analytics on the dashboard.

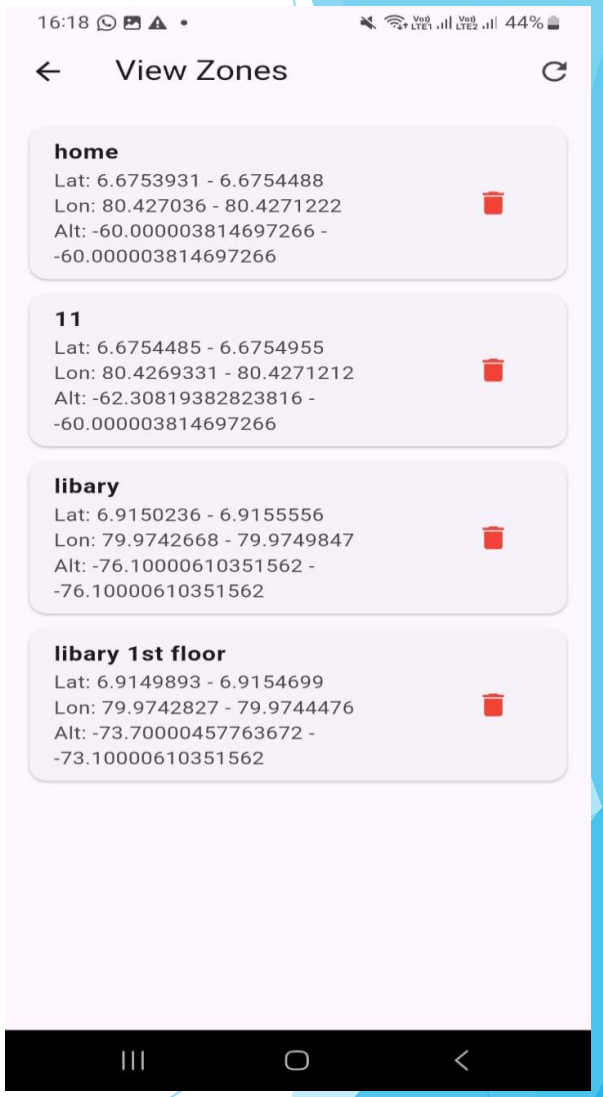

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46%
Welcome
Choose how you want to continue
Client
Find products, try outfits, and get recommendations
Store
Manage zones, products, and store information


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Store Dashboard
Create Zone
Create a zone using 4 location points
Identify Zone
Check which zone a location belongs to
View Zones
View and delete stored zones

GPS-Based Customer Behaviour Tracking & Real-Time Staff Assistance







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User Requirements Addressed

- ✓ Real-time customer monitoring
- ✓ Easy zone management
- ✓ Automatic behaviour detection
- ✓ Faster staff response
- ✓ Improved customer support
- ✓ Low-cost & scalable solution

Design Excellence & User Feedback

Design Excellence

- Clean, modern GPS-based store map interface
- Real time customer tracking inside store zones
- Fast and easy notification system
- Minimal and user friendly UI for quick decision

User Feedback

- Real-time tracking and alerts are very useful
- Helps staff respond faster to customers
- Managers like behaviour analytics dashboard
- Needs better GPS accuracy indoors
- Clearer priority indicators suggested

Future Works

- Improve zone creation system (done)
- Improve GPS zone detection (done)
- Real-time customer behaviour tracking
- Staff alert system
- Store analytics dashboard

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AI-Based Virtual Try-On and Size Recommendation System

Problem Definition & Research Gap

Research Gap

Existing fashion systems rely on manual measurements or 3D scanning and lack accurate integration of virtual try-on with size prediction, leading to poor fit accuracy.

Novelty

Combines Gemini-based virtual try-on with ML-based size prediction to give accurate fit and visualization without 3D scanning or manual measurements.

Solution Overview

Problem

- Customers cannot try clothes online
- Wrong size selection issues

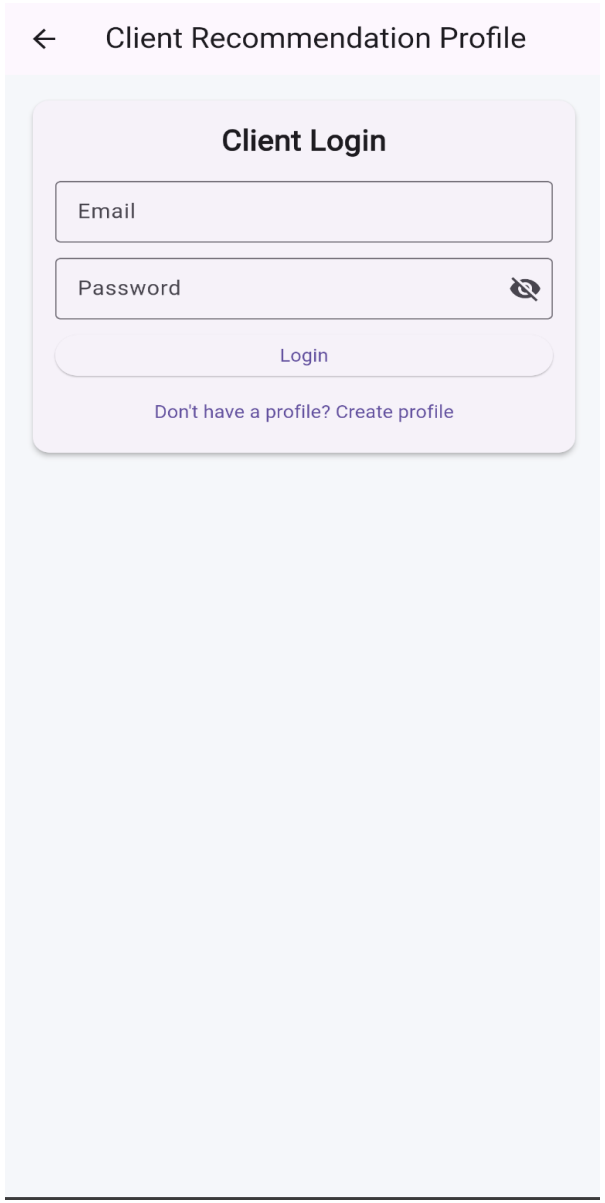
Solution

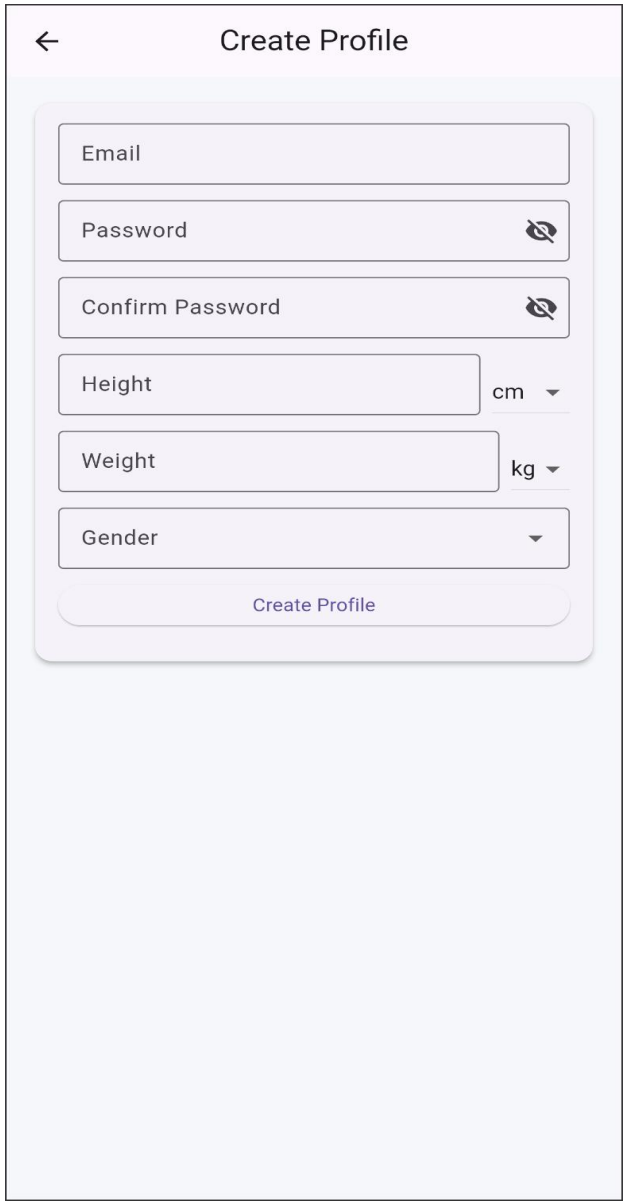
- AI generates virtual try-on results
- ML predicts accurate clothing sizes

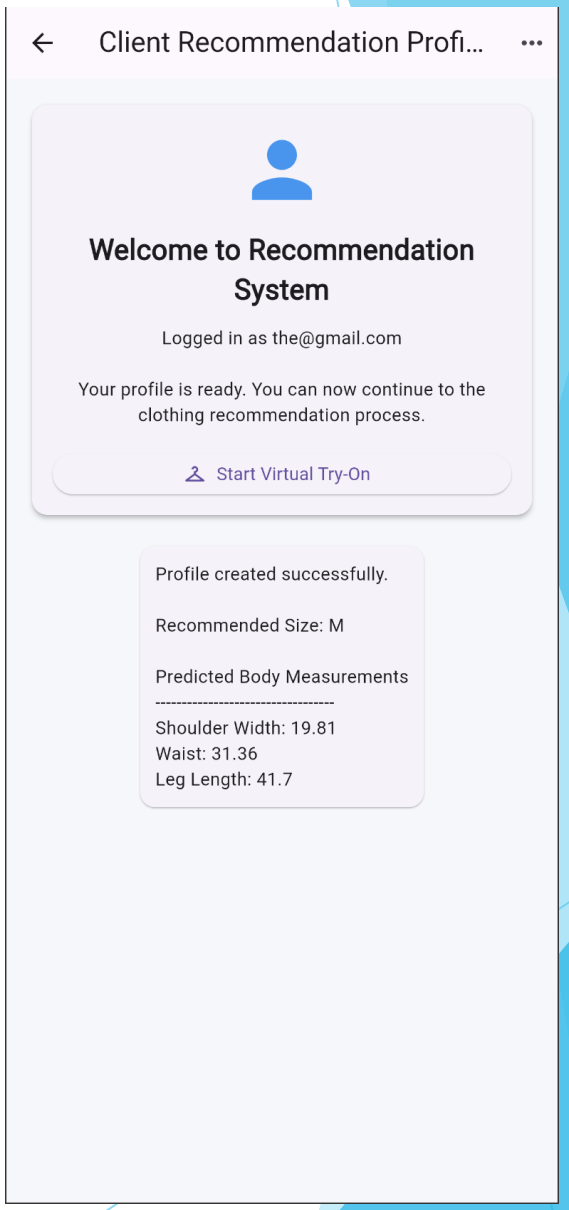
Workflow & Prototype

- Upload human and cloth images
- Gemini API generates try-on output
- User enters height and weight
- ML predicts Size, Waist, Shoulder Width, Leg Length

AI-Based Virtual Try-On and Size Recommendation System



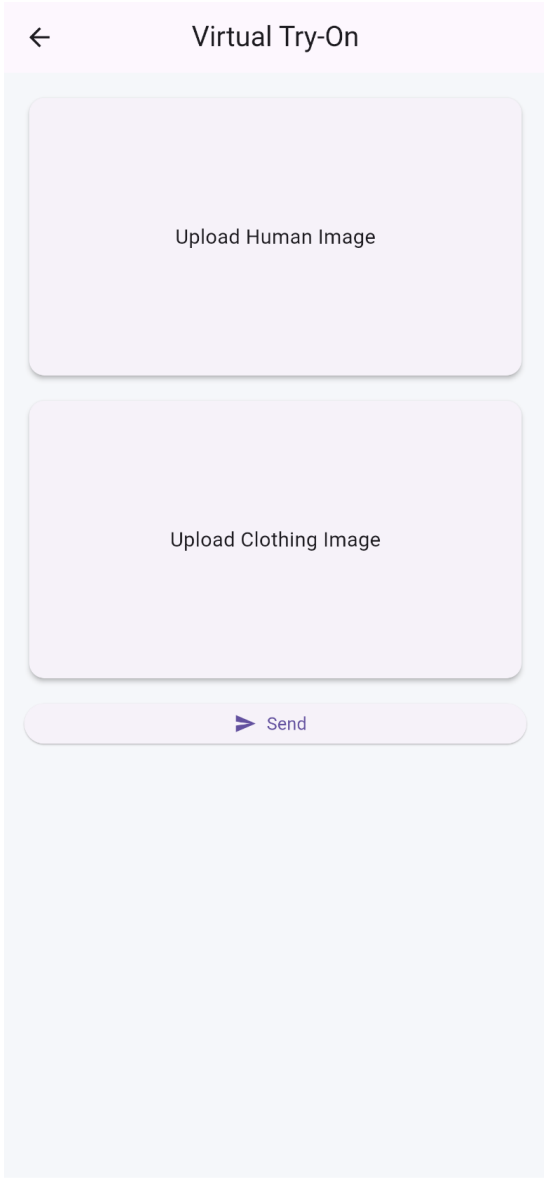


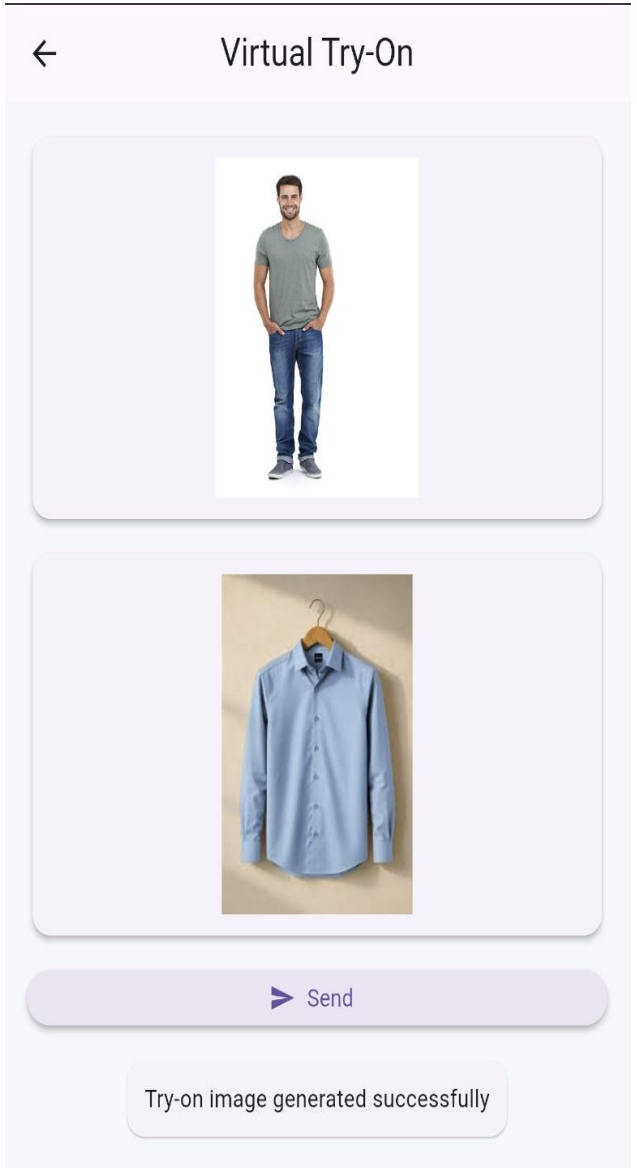


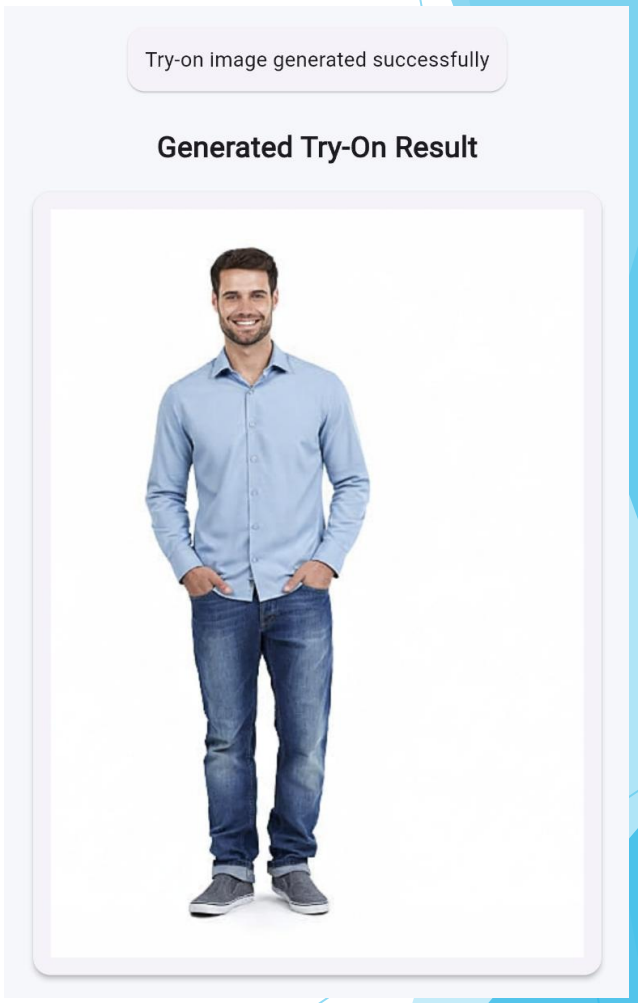
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AI-Based Virtual Try-On and Size Recommendation System







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User Feedback on Prototype

- Users found the virtual try-on feature helpful for visualizing outfits before purchase
- Positive feedback on the simplicity and ease of profile creation
- Users appreciated personalized size recommendations
- Prototype improved confidence in online clothing selection
- Some users suggested improving image quality and response speed
- Additional feedback requested more clothing categories and better UI design

Design Excellence / Contribution

- Combines **Generative AI** and **Machine Learning** into a single smart fashion solution
- Uses **Gemini API** for realistic virtual try-on image generation
- Reduces size mismatch and improves online shopping confidence
- Simple and user-friendly system without requiring expensive 3D scanning technology

Future Works

- Improve **user experience** to make the system more easy, smooth, and helpful
- Complete **frontend development** and improve UI/UX design for better usability
- Backend development is currently **75% completed**, needs final integration and optimization
- Further **training of the ML model** with more data to improve size prediction accuracy
- Ongoing **testing and performance optimization** to improve system reliability

IT22244598 | Methsara W.M

Specialization: Information Technology



Smart Inventory & Stock Flow Optimization System

AI-Powered Retail Inventory Intelligence for Omni-Retail Environments

Problem Definition & Research Gap

Research Gap

Existing retail inventory systems mainly focus on stock tracking and lack AI-based demand forecasting and intelligent inventory decision support.

Novelty

This research introduces an AI-powered inventory optimization system that combines demand forecasting, inventory analysis, and AI-generated business insights to improve retail stock management.

Solution Overview

Problem

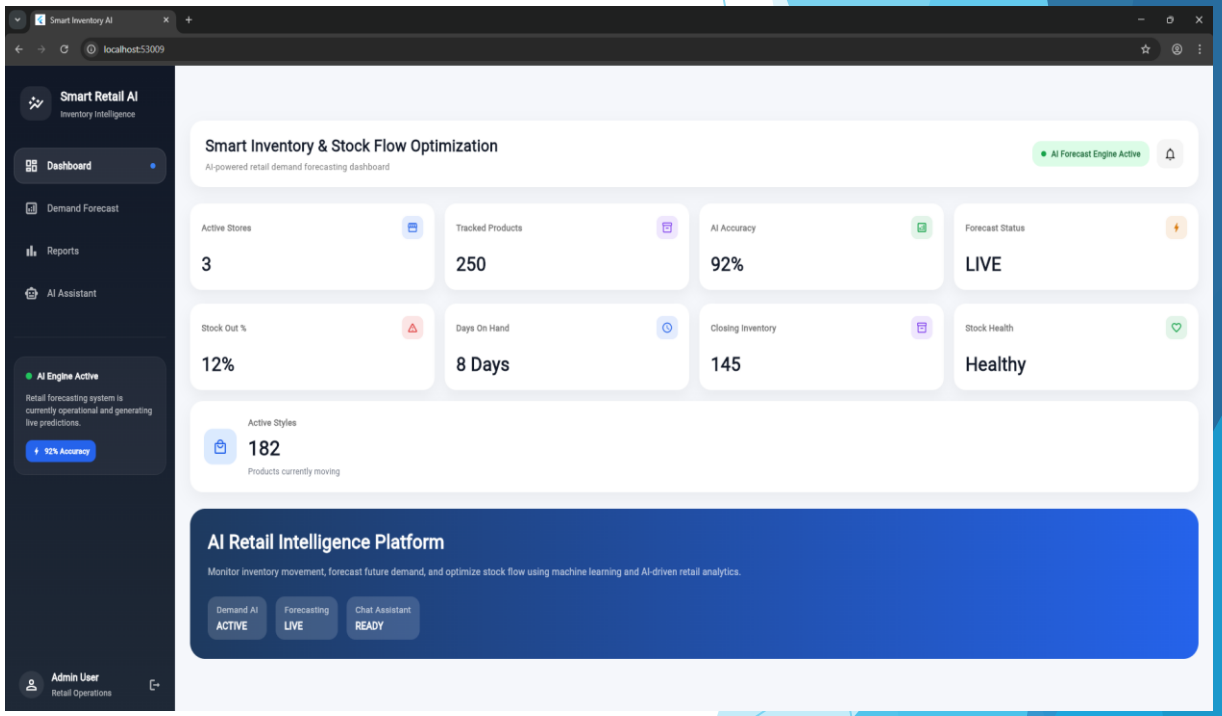
- Retail stores face stock shortages and overstock situations
- Limited intelligent inventory decision support
- Poor visibility of stock movement and demand trends
- Difficulty predicting future inventory demand accurately

Solution

- AI-based demand forecasting system
- Inventory optimization and stock recommendation engine
- AI-generated business insights and explanations
- Conversational AI retail assistant for inventory analysis

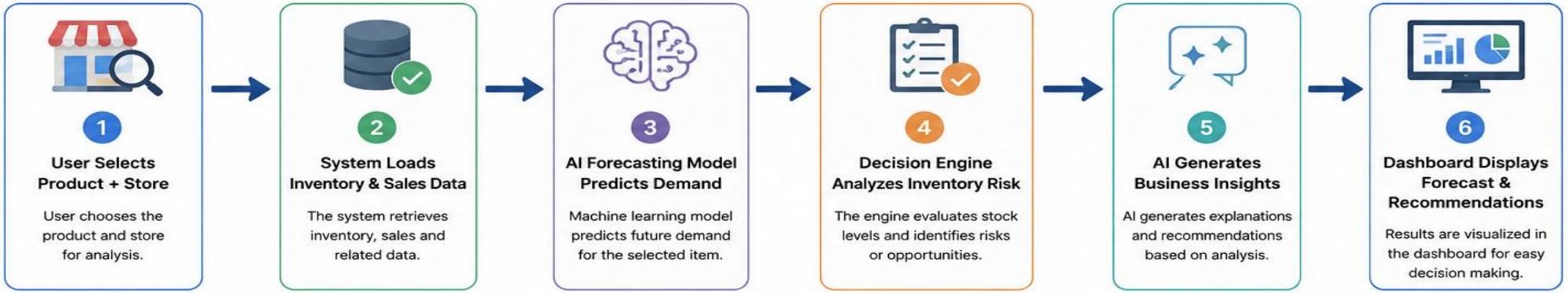
Technologies

- Flutter, FastAPI, XGBoost, Gemini AI



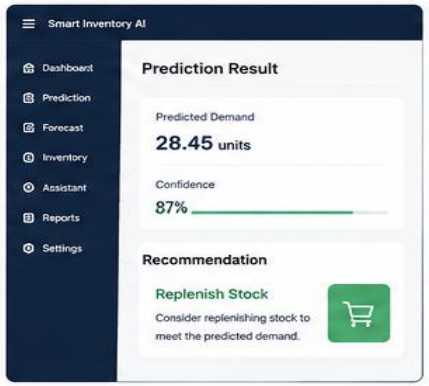
Workflow & Prototype

System Workflow



Prototype Features

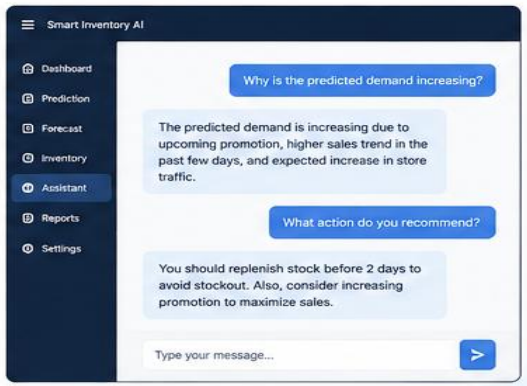
- AI Demand Forecasting
- 7-Day Forecast Visualization
- Inventory Recommendation Engine
- AI Business Explanations
- Conversational AI Retail Assistant
- KPI Analytics Dashboard



Prediction Page

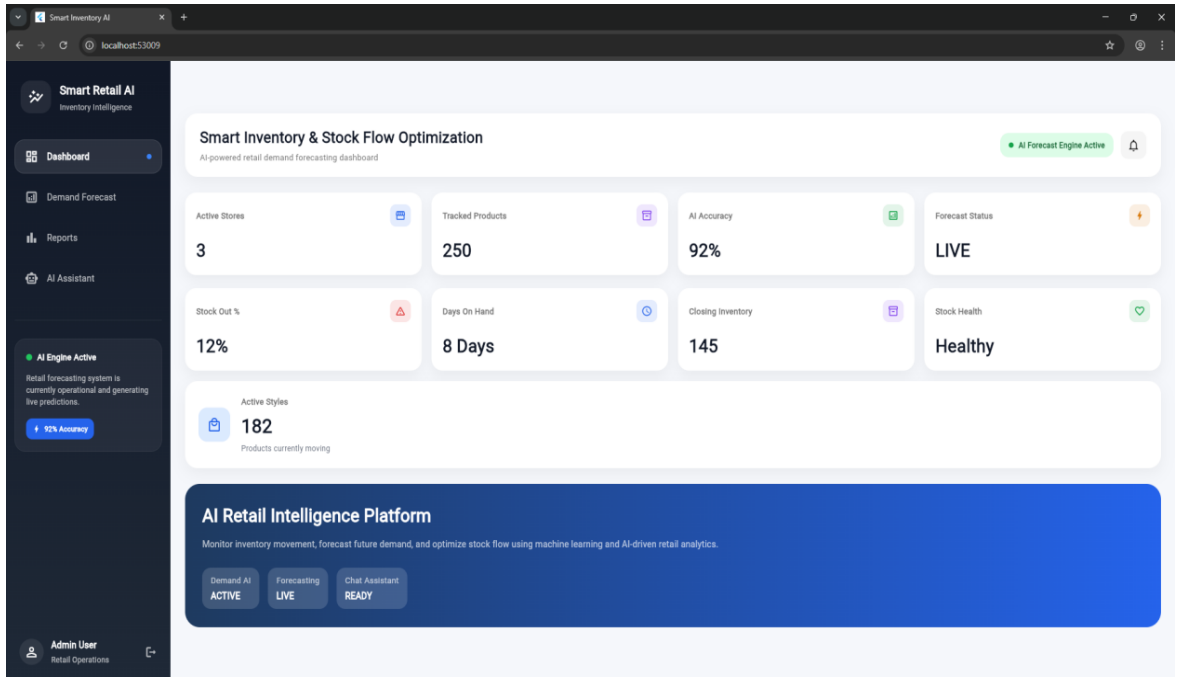


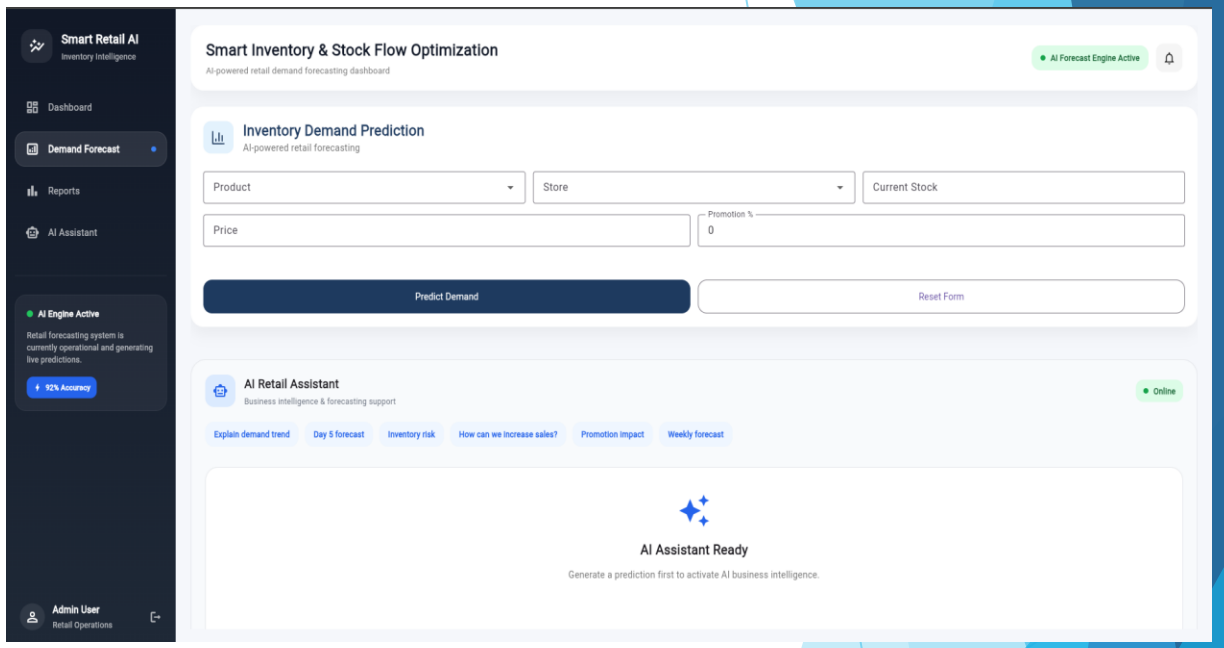
7-Day Forecast



AI Assistant

Smart Inventory & Stock Flow Optimization System

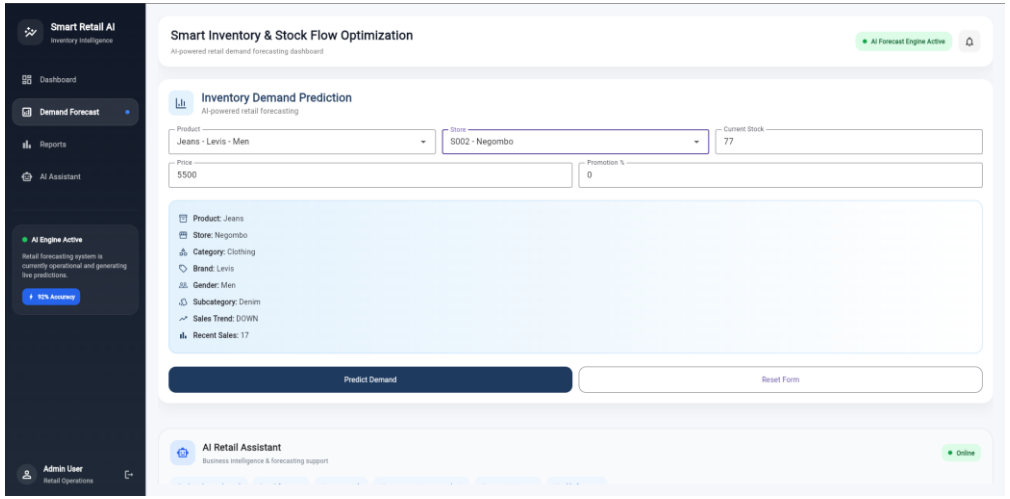


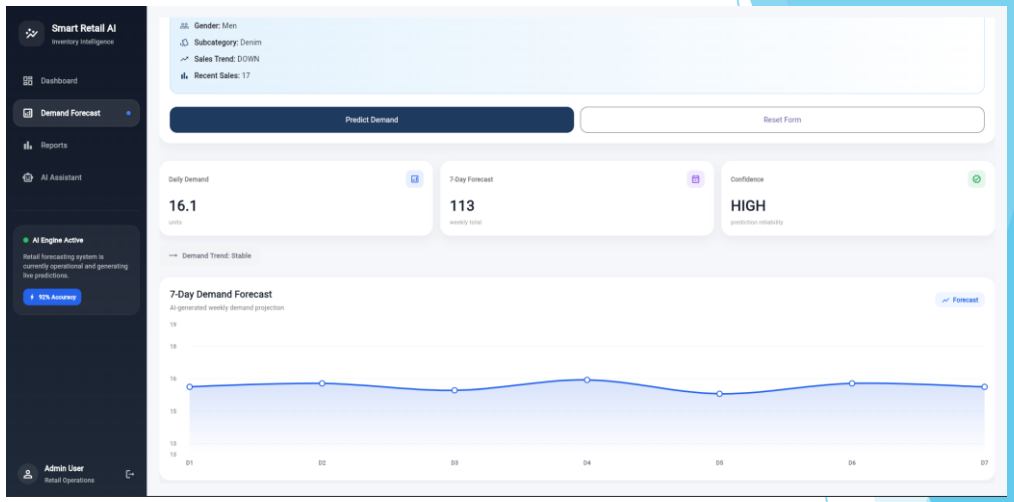


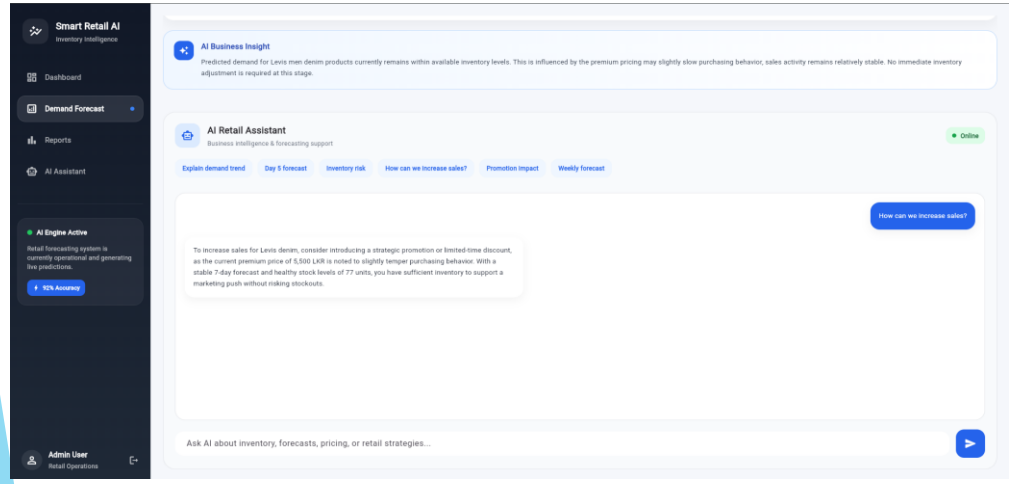
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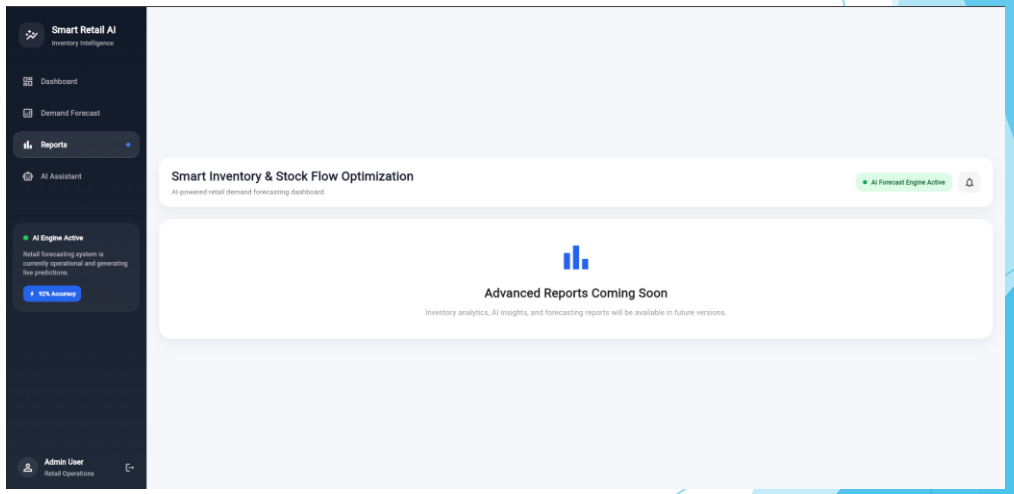
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Smart Inventory & Stock Flow Optimization System









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User Requirements Addressed by the Solution

Functional Requirements

- ✓ Product and store selection
- ✓ AI demand forecasting
- ✓ Forecast visualization dashboard
- ✓ Inventory monitoring
- ✓ AI-generated inventory recommendations
- ✓ Conversational AI assistant

Business Requirements

- ✓ Reduce stock shortages
- ✓ Improve inventory visibility
- ✓ Support retail operational decisions
- ✓ Improve stock movement efficiency

Non-Functional Requirements

- ✓ User-friendly interface
- ✓ Fast prediction response time
- ✓ Responsive dashboard design
- ✓ Scalable backend architecture
- ✓ Reliable AI prediction workflow

Design Excellence & User Feedback

AI-Powered Architecture

- ✓ Integrated forecasting, inventory logic & AI reasoning
- ✓ Hybrid AI explanation: local logic + Gemini AI
- ✓ Modular backend service architecture

Modern Dashboard Design

- ✓ Enterprise-style analytics dashboard
- ✓ KPI overview cards & forecast trend visualization
- ✓ AI retail assistant integration

Fashion Retail Intelligence Features

- ✓ Brand & gender-based inventory segmentation
- ✓ Promotion impact & forecast confidence analysis
- ✓ Inventory optimization recommendations

Technical Contribution

- ✓ Flutter + FastAPI integrated architecture
- ✓ AI-driven inventory forecasting workflow
- ✓ Forecast enhancement & AI explanation engine

User Feedback on Prototype

Positive Feedback

- ✓ Modern and professional UI
- ✓ Forecasts easy to understand
- ✓ Useful AI-generated insights
- ✓ Interactive AI assistant
- ✓ Helpful inventory recommendations

Suggestions for Improvement

- ✓ Monthly forecasting
- ✓ Stock health analytics
- ✓ Slow/fast moving analysis
- ✓ Advanced retail intelligence metrics

Future Works - Planned Enhancements

Advanced Forecasting

- ✓ Monthly and quarterly demand forecasting
- ✓ Dynamic custom time-period predictions
- ✓ Custom date range demand forecasting

Inventory Intelligence

- ✓ Slow-moving and fast-moving item analysis
- ✓ Advanced stock health monitoring
- ✓ Days-on-hand inventory analytics

AI & Business Insights

- ✓ Enhanced AI-generated retail recommendations
- ✓ Smarter conversational retail assistant
- ✓ Advanced promotion impact analysis

System Expansion

- ✓ Real-time database integration
- ✓ Multi-branch inventory synchronization
- ✓ Scalable enterprise deployment architecture





IT22146588 | WIJETUNGA J A R K

Personalized Marketing Intelligence

Contribution : Marketing Intelligence Dashboard

An intelligence marketing dashboard which analyses customer emotions , behaviors and behaviors and help stakeholders to gain insight for better decision making

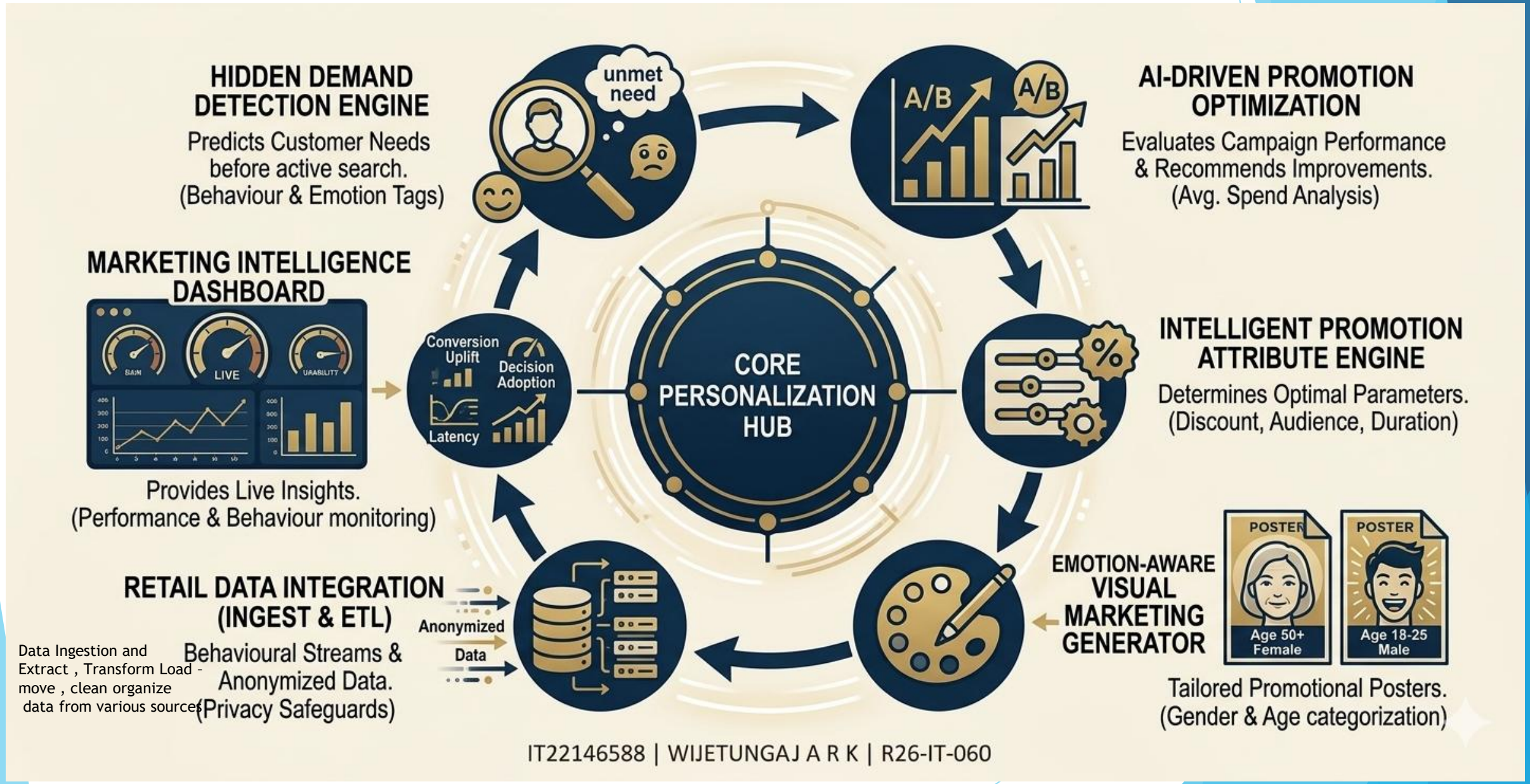
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Problems identified and the proposed solutions (proto type)

No	Problem	Proposed proto type solution
	Generic promotions	Random Forest recommender predicts best offer type per customer segment
2	Lack of insights into promotion performance and customer responses	Promotion Performance tab shows CTR, redemption rates, revenue lift
3	Low customer engagement due to poorly targeted promotions	Customized poster generator creates targeted visuals per age/gender
4	Limited understanding of individual customer preferences and purchasing behavior	Promo Attributes tab analyses preferences by age group, gender, loyalty tier
5	Inability to identify hidden or future customer demand	Random Forest predicts which promotions customers will respond to before campaigns run
6	Manual marketing campaign creation which reduces efficiency	Automated poster generation and batch segment recommendations reduce manual effort

Personalized Marketing Intelligence

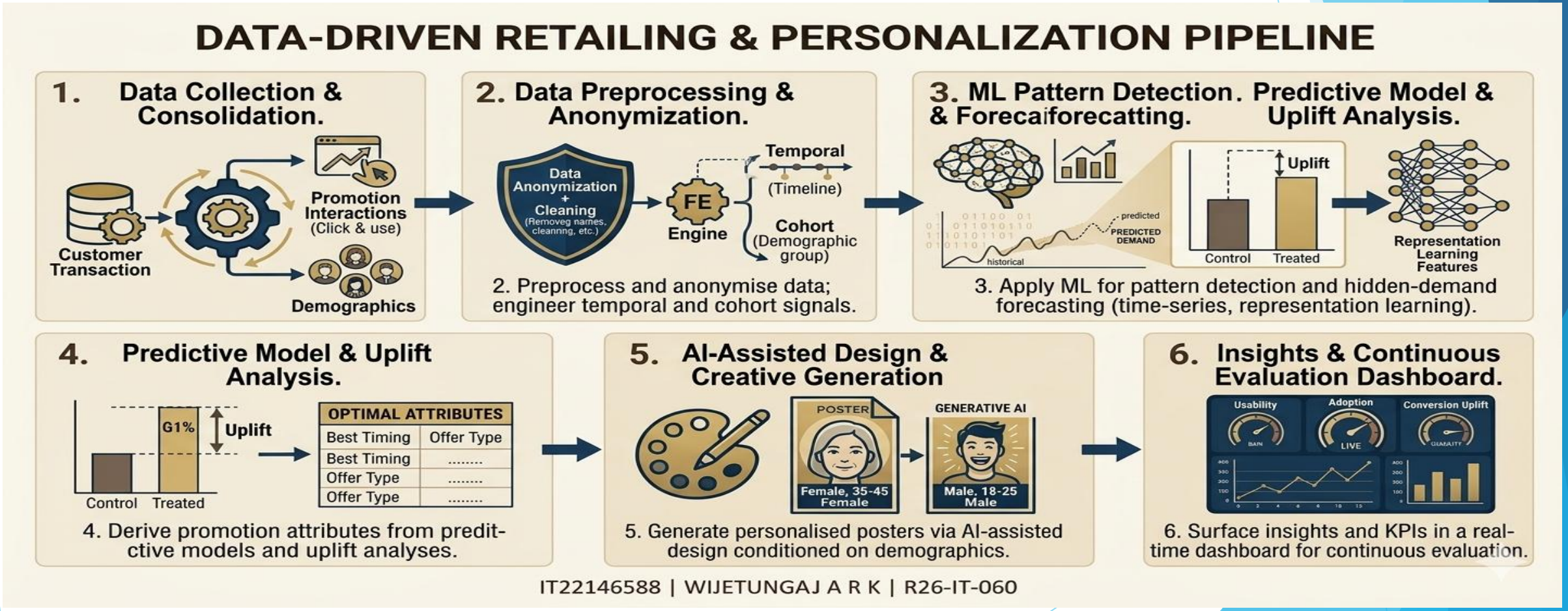
Novelties :



Data Ingestion and Extract , Transform Load – move , clean organize data from various sources

Application of Domain Knowledge & Methodology

Domain knowledge spans retail marketing, customer behaviour analysis, machine learning segmentation, transaction analytics and marketing design intelligence for emotion-intelligence for emotion-aware creative outputs.



Design Excellence / Contribution

Novel System Combination :
First system to combine Random Forest promotion recommendation + emotion-aware poster generation in a single retail dashboard
Tools: *Python*, *Streamlit*, Random Forest , *Stable Diffusion v1.5*

End-to-End Pipeline
Single workflow: CSV upload → data cleaning → analytics → Random Forest prediction → poster generation → download
No separate tools needed — everything in one interface
Tools: Streamlit, Pandas, Scikit-learn, Random Forest , Stable Diffusion, Matplotlib

Interactive Filtering & Segmentation
Dynamic filters for offer type, channel, age group and gender across all analytics tabs
Batch prediction generates recommendations for all demographic combinations and exports as CSV
Tools: Streamlit widgets, Pandas, Random Forest

Visual Generation – Poster Generation
Promotional posters generated using Stable Diffusion v1.5 running on Google Colab GPU
Psychographic tone mapping selects visual mood based on customer segment
Example: "elegant, sophisticated" for Female 26-34 vs "bold, dynamic" for Male 18-25
Tools: Stable Diffusion v1.5, Flask API, Google Colab, ngrok, Pillow

User Requirements Addressed:

Functional requirements addressed:

•Upload and clean campaign and customer data automatically
•View promotion performance analytics with filters
•Get AI-powered promotion type recommendations per customer segment
•Generate personalised promotional posters based on demographics
•Download batch promotion recommendations as CSV
•View feature importance to understand model decisions

Non-functional requirements addressed:

•Privacy — PII (Personally Identifiable Information)columns automatically removed on upload
•Usability — simple tab-based navigation, no technical knowledge required
•Accuracy — Random Forest model trained to be 80%+ accuracy on realistic retail data

User Feedback on Prototype :

1. Improve on the image of the post and the wordings to be clear and sensible
2. To diversify and add more details to the dashboard to give a broader view of the promotion scale

Future Work

•**Firebase/Cloud Database Integration** — store campaign results, predictions and generated posters in a cloud database for multi-user access and historical tracking

•**Email Campaign Integration** — directly send generated posters to the recommended customer segments via email without leaving the dashboard

•**Improved Image Quality** — upgrade from Stable Diffusion v1.5 to DALL-E 3 or Stable Diffusion XL for higher quality, more realistic promotional posters

•**A/B Testing Module** — automatically run and evaluate two promotion variants against each other and select the better performing one

•**Customer Sentiment Analysis** — analyse customer reviews and social media mentions to detect promotion fatigue and adjust recommendations accordingly

•**Expanded Model Training** — retrain XGBoost on ***larger real retail datasets*** to improve prediction accuracy beyond the current simulated data

•**Role-Based Access Control** — different dashboard views for store managers, marketing teams and senior management with appropriate data access levels

Commercialization & Sustainability of the Solution

Customer Persona

- Retail store owners and managers who want to improve customer experience, optimize inventory, and increase sales using data-driven insights.

Market Size

- The solution targets small to medium retail businesses and growing retail chains that are adopting digital transformation and AI-based decision systems.

What is Unique in Our Solution

- Integrates **customer behavior, virtual try-on, inventory, and marketing** in one platform
- Uses **AI + Generative AI** for smarter retail decisions
- **Low-cost and scalable** for real-world retail environments

Commercialization & Sustainability of the Solution

How Will the Cost Be Recovered

- Subscription-based model (monthly/annual)
- Retailers pay for:
  - analytics dashboard
  - AI insights
  - recommendation services
- Additional revenue through **premium features and system customization**

Target Market

- Retail stores
- Fashion outlets
- Shopping malls

Revenue Model

- SaaS Subscription
- Service-based pricing

Value Proposition

- Better decisions
- Increased sales
- Improved customer experience